

PULSE OXIMETER LESSON PLAN

Grade: 6-8

Topic: Pulse Oximeter Setup and Coding

Lesson Focus and Goals:

- 1) Identify basic coding building blocks (variables, data types, loops, and functions) used in Arduino programs.
- 2) Predict and fill in missing code that turns LEDs on/off and reads values from a light sensor.

Materials Needed:

- 1) Arduino Uno
- 2) Bread Board
- 3) USB cables
- 4) Red LED
- 5) IR LED
- 6) Photodiode
- 7) 3 Resistors
- 8) Wires
- 9) Projector + Powerpoint
- 10) Printed worksheets with fill-in-the-blank coding exercises
- 11) Example Arduino code with parts missing

Key Points That Must Be Covered:

- 1) What a variable is and why data types matter ("int", "float", "bool")
- 2) What input (sensor readings) and output (LED control) mean in an Arduino program
- 3) The purpose of setup() and loop() in Arduino sketches
- 4) How to turn an LED ON/OFF using digitalWrite()
- 5) How to read a sensor value using analogRead()
- 6) How to use the Serial Monitor to print data
- 7) How to fill in missing code in a guided program

Activities (with timing):

- 1) (5 min) Warm-Up
 - a) Show a pulse oximeter or picture
 - b) Ask “How do you think this device measures your pulse with LIGHT?”
- 2) (7 min) Mini Lecture on Variables & Data Types
 - a) Explain variables as "buckets" holding information
 - b) Teach what int, float, and bool are and give examples + analogies

- i) int can be thought of as short for “integer” - A whole number (no decimals or fractions)
 - (1) An int is like a jar that can hold only whole candies, nothing cut in half
 - (2) Examples: 12, 0, -5
 - (3) When we use it: Pin numbers on a circuit (int RED_LED = 12;), counting things, storing whole-number
 - ii) float is short for “floating-point number” - A number with decimals, more precise values
 - (1) A float is like a money, that lets you have things like cents, a nickel = 0.05 dollars
 - (2) Examples: 3.14, 0.75, 98.6
 - (3) When we use it: Calculations that need decimals, measurements (temperature, oxygen levels, averages), anything that isn’t a whole number
 - iii) bool is short for “boolean” - Only true or false
 - (1) A bool is like a light switch, it can only be ON or OFF, nothing in between
 - (2) Example: bool isLightOn = true;
 - (3) When we use it: Checking if something is ON or OFF, storing yes/no decisions
- 3) (4 min) – Activity: Students identify which data type is appropriate when shown 3 short examples
 - a) Example 1 — LED Pin Number
 - i) LED_PIN = 9;
 - ii) Correct answer: int
 - iii) Why? → LED pins are whole numbers → integers
 - b) Example 2 — Sensor Reading With Decimals
 - i) temperature = 23.7;
 - ii) Correct answer: float
 - iii) Why? → Temperatures often use decimal values → floating point
 - c) Example 3 — LED On or Off
 - i) buttonPressed = false;
 - ii) Correct answer: bool
 - iii) Why? → A boolean stores true or false
- 4) (8) min – Mini Lecture + Demo: Inputs and Outputs
 - a) Explain digital outputs → LEDs
 - b) Explain analog inputs → reading the light sensor
 - c) Demo turning an LED on/off with simple code
 - d) Demo reading a sensor and printing to Serial Monitor

- 5) (15 min) – Guided Hands-On Worksheet
 - a) Students complete a fill-in-the-blank Arduino program
 - b) Examples they fill in:
 - i) `_____ RED_LED = 12; // choose correct data type`
 - ii) `pinMode(RED_LED, _____); // input or output?`
 - iii) `digitalWrite(RED_LED, _____); // turn LED on`
 - iv) `int reading = analogRead(_____); // which pin?`
 - v) `Serial.println(_____); // print value`
 - vi) `digitalWrite(RED_LED, _____); // turn it off`
- 6) (10 min) – Review
 - a) Students share their filled-in code and we type it into Arduino
 - b) Test by:
 - i) Watching the LED turn on/off
 - ii) Observing sensor readings change on the Serial Monitor
 - iii) Placing a finger over the ALS to see the numbers change
- 7) (5 min) – Implementation into PulseOx code
 - a) Share screen with our official commented out Arduino code with missing spaces
 - b) Students volunteer the answers for the missing code
 - i) choosing the correct data type for a variable
 - ii) completing a `pinMode()` line
 - iii) writing a `digitalWrite(LED, HIGH)` or `LOW`
 - iv) completing an `analogRead()` line
 - v) filling in a `Serial.print()` or `Serial.println()` call
 - c) We type answer into the official program
 - d) Demonstrate the fully functioning PulseOx sampler so they see how their basic building blocks support a real project

How This Lesson Transitions in and Out

- 1) Start Transition: Begin with a real-world device (pulse oximeter) → connect it to lesson purpose → bridge into variables and sensors.
- 2) Middle Transition: Move from slides → demonstration → hands-on coding by saying “Now that you know how Arduino reads and prints sensor values, you’ll help build part of the program yourself.”
- 3) End Transition: Wrap-up with a final demonstration using their implemented code

How Student Understanding Will Be Gauged:

- 1) Poll questions (during early lecture)
- 2) Asking students to justify data-type choices
- 3) Observing worksheet completion accuracy
- 4) Debugging during upload/test

5) Final questions